

Ticket Show App Project Report:

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Hello, I am Rajat Kumar, currently pursuing B.S. in Data Science and Programming from IIT, Madras along with that I am also pursuing B. Tech from MIT, Muzaffarpur in Electrical Engineering Stream. Here I am in 4th year.

Description

My project is a ticket show app which is made with the motive to provide easy ticket booking system for their favourite show. Also, An Admin can host their venue and add show for people to book its ticket. An admin can also manage the venue and show anytime. It can be used for booking show tickets and users can book many tickets for many movies. Also An admin can get the statical summary via chats. I have also implemented crud operations on admin, user, show and venue using API.

Technologies used.

1. Python Language, Template Engine – Jinja2 (for creating dynamic webpages), HTML5, CSS (for Building and styling web pages)
2. Web Framework – Flask, Flask login, Flask CORs (for Cross-origin resource sharing in Flask), render_template, request, redirect, url_for , Flask flash (for showing feedback messages to users in Flask, Flask Restful (Python framework for building RESTful APIs)
3. Database – SQLite (for relational database management system), Flask SQL Alchemy (for Object-relational mapping in Flask), Matplotlib (for creating data visualizations in Python), uuid (for generating unique identifiers), Datetime (for working with date and time objects like getting current time)

DB Schema Design

Tables	Columns	Type	Relationship	Related Tables
User	id, username, password	Model	-	-
Admin	id, username, password	Model	-	-
Venue	v_id, v_name, v_location, v_capacity, v_admin_id	Model	admin	Admin
Show	s_id, s_name, s_rating, s_timing, s_tags, s_price, s_director, s_capacity, s_available_seats, s_admin_id	Model	admin	Admin
Booking	id, user_id, show_id, venue_id, date	Model	user, show, venue	User, Show, Venue
Rate	id, rating, user_id, show_id	Model	user, show	User, Show
show_venue	venue_id, show_id	Association	show, venue	Show, Venue
show_user	sh_id, user_id, show_id	Association	user, show	User, Show
venue_user	ve_id, user_id, venue_id	Association	user, venue	User, Venue

Relationship Type	Table 1	Table 2
One-to-Many	User	Booking
One-to-Many	Admin	Venue
One-to-Many	Admin	Show
Many-to-Many	Show	Venue
Many-to-Many	User	Show
Many-to-Many	User	Venue
One-to-One	User	Rate
One-to-One	Show	Rate

API Design

I can get details of each admin and the information about the venue and show he/she has created. Also, for users I can get information regarding them and their bookings. I can perform CRUD operation on show and venue i.e., creating a venue/show, reading, updating, and deleting a particular show/venue. I have only written code for these operations and illustrate it on postman or thunder client.

Architecture and Features

I have used different files for controllers, templates, and model. In instance folder, db file is located. Controllers are in app.py file where I have imported necessary modules and connected db and api and initialised it. After that I have defined routes and functions for user/admin login/signup, venue management routes, show management, booking managements, ratings and lastly the summary part code where I have generated charts. At last, it consists of app.run code.

In templates I have stored all html files and in static file, I have stored CSS and image files. In resources.py file, I have written the code the api crud operations. Requirements.txt file contains the names of all installed packages and README.md file is also there for easy implementation of applications.

Features:

1. Multiple admin/user functionality, user can book tickets of multiple show at multiple venue.
2. User can search all shows based on the show name, venue name and genre category.
3. User can rate the shows which they have booked. User can see their bookings. User can only book tickets if seats are available, user can see total price of their tickets.
4. Admin can do crud operation on venue and show.
5. Admin can see the statical summary of venue vs no. of show hosted, venue vs no. of bookings, all show overall rating and booking chart and charts of no. of show of particular genre being hosted by a venue and its booking.

Additional Features:

1. **API** for crud operation on venue and show, get all information about admin and user.
2. Backend validation and flash message on invalid in inputs, Styling and Aesthetics

Video

<https://drive.google.com/file/d/1GqoqPb8xM5ufHA2u8-e5POcoQISY7HMz/view?usp=sharing>